2026 American Control Conference (ACC)
May 27-29, 2026. New Orleans, Louisiana, USA

# Static Friction Modeling and Compensation for Improved Force Tracking in Pneumatic Cylinders Using Sliding Mode Control

Ammar Yacoub[1], Edmond Richer[2], Yildirim Hurmuzlu[3]

*Abstract*— **Static friction, also known as stiction, is a primary source of tracking error in pneumatic cylinders, which significantly limits their use in high-precision force control tasks. This paper presents a detailed experimental study of the static friction behavior of a small pneumatic cylinder, examining how the friction force depends on piston position and supply pressure. Based on these findings, a new static friction model that considers pressure and position is developed to accurately represent the stiction phenomenon. Additionally, a Sliding Mode Controller (SMC) is designed and implemented to produce a desired output force. The SMC's performance is evaluated experimentally both with and without the proposed friction compensation model. Results show that the SMC with the integrated compensation reduces steady-state force tracking errors compared to the uncompensated controller.**

## I. Introduction

A haptic interface is a device that enables communication between humans and machines by providing users with sensations of contact and reaction forces. This allows them to manipulate objects in a virtual environment or remote setting [1], [2]. This technology is essential for bridging the sensory gap caused by physical distance or virtual interactions within virtual environments [3].

Over the past several years, various haptic interfaces have been developed for multiple applications, including virtual-reality surgical training [4], robot-assisted surgery [5], teleoperation [6], [7], and rehabilitation [8]. These interfaces vary from simple single-DOF devices [9] to advanced multi-DOF devices [6], [10].

Haptic interfaces are divided into grounded, handheld, and wearable devices based on their level of wearability [11]. The most common wearable haptic devices are haptic gloves and exoskeleton systems, which are designed to provide kinesthetic haptic feedback when worn on the user's body [12].

Haptic gloves utilize three types of actuators: electric motors [13], pneumatic actuators [14], [15], and shape memory alloy (SMA) actuators [16]. Pneumatic actuators produce motion through compressed air and offer a high power-to-weight ratio, along with many advantages such as cleanliness, easy maintenance, mechanical simplicity, and low cost [17].

However, friction is always present in actuators, such as permanent magnet linear motors [18] and pneumatic cylinders, mainly between piston/rod and contact surfaces. It can occur nonlinearly, leading to limit cycles and unexpected stick-slip oscillations at low operating velocities [19] that make it difficult to generate the precise feedback force needed for a haptic glove. Therefore, understanding the friction behaviors of pneumatic cylinders and developing an appropriate friction model are essential to improve force control accuracy [20].

Several experimental studies have been proposed to investigate the friction behavior in pneumatic cylinders [17], [21]. However, most of these studies focus on the friction of large cylinders. This paper, however, introduces an experimental setup to model the static friction behavior of small pneumatic cylinders as a function of supply pressure and piston position, and demonstrates the ability of a sliding mode controller to track desired force with friction compensation.

This paper is organized as follows: Section II provides a brief overview of pneumatic cylinder force modeling. Section III describes the design of the sliding-mode controller (SMC). Section IV presents the experimental setup and results. Finally, Section V provides the conclusion and future work.

## II. Pneumatic Cylinder Force Modeling

Figure 1 shows the schematic of a pneumatic system, which consists of a double-acting pneumatic cylinder actuated by a 5/3 proportional valve. The dynamic model of this system is detailed in [22]. The key features of the dynamic model are summarized in this paper to support the development of the sliding-mode controller with friction compensation. The force produced by the actuator shown is

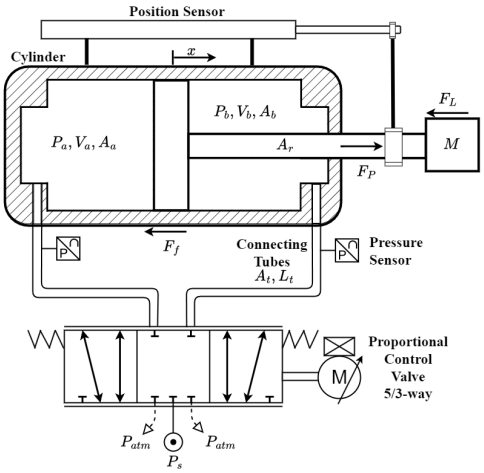



Fig. 1: Schematic representation of the pneumatic system.

[1]Ammar Yacoub, and Yildirim Hurmuzlu[3] are with the Systems Laboratory, Southern Methodist University, Dallas, TX 75205 USA (e-mail:ayacoub@smu.edu, hurmuzlu@lyle.smu.edu)

[2]Edmond Richer is with the Biomedical Instrumentation and Robotics Laboratory, Southern Methodist University, Dallas, TX 75205 USA (e-mail:richer@lyle.smu.edu)

979-8-3315-9381-0//$31.00 ©2026 AACC


Authorized licensed use limited to: Southern Methodist University. Downloaded on October 18,2026 at 21:43:04 UTC from IEEE Xplore. Restrictions apply.

given by:

$$F_p = P_a A_a - P_b A_b - P_{atm} A_r \tag{1}$$

where $P_a$ and $P_b$ are the absolute pressures inside each chamber, $A_a$ and $A_b$ are the effective areas of each chamber, and $A_r$ is the cross-sectional area of the piston rod.

The pressure dynamics in the two chambers, $\dot{P}_a$, and $\dot{P}_b$, are governed by Eq. 2. This model assumes that the air behaves as an ideal gas, with pressure and temperature evenly distributed in each chamber. It also considers the kinetic and potential energies of the air to be negligible and assumes that the charging and discharging processes occur at constant temperature (isothermal).

$$\dot{P}_a = \frac{RT}{V_a}\dot{m}_a - \frac{P_a}{L_a + x}\dot{x}, \quad \dot{P}_b = \frac{RT}{V_b}\dot{m}_b + \frac{P_b}{L_b - x}\dot{x} \tag{2}$$

where $R$ is the ideal gas constant, $T$ is the absolute temperature, $L_a$ and $L_b$ are the chambers' lengths, $\dot{m}_a$ and $\dot{m}_b$ are the mass flow rate in or out of the chamber, and $V_a$ and $V_b$ are the chamber's volume, which depends on the piston position given by $x$.

### A. Air Flow Through Proportional Valves

The mass flow rate ($\dot{m}_{a/b}$), can be expressed algebraically in relation to the valve area, as given by Eq. 3. This relies on the assumption that the flow through the valve acts as an ideal gas undergoing an isentropic process, leading to the commonly used formula for mass flow rate expression [22].

$$\dot{m}(P_u, P_d, z, C_f) = C_f A_v(z) \psi(P_u, P_d) \tag{3}$$

where $C_f$ is the non-dimensional discharge coefficient of the valve, reflecting the precise geometry of the orifice being modeled and requiring experimental identification. $A_v(z)$ denotes the valve's aperture area (orifice area), which depends on the spool position ($z$) and must be determined experimentally. $\psi(P_u, P_d)$ is expressed as:

$$\psi(P_u, P_d) = \begin{cases} C_1 \frac{P_d}{\sqrt{T}} & , \frac{P_d}{P_u} \le P_{cr} \\ C_2 \frac{P_d}{\sqrt{T}} \left(\frac{P_d}{P_u}\right)^{\frac{1}{\gamma}} \sqrt{1 - \left(\frac{P_d}{P_u}\right)^{\frac{\gamma-1}{\gamma}}} & , \frac{P_d}{P_u} > P_{cr} \end{cases} \tag{4}$$

where $\gamma$ is the ratio of the fluid's specific heat capacity at constant pressure ($C_p$) and constant volume ($C_v$).

The critical pressure ratio $P_{cr}$ distinguishes between subsonic (unchoked) and sonic (choked) flow through the orifice and is expressed in terms of $\gamma$ as

$$P_{cr} = \left(\frac{2}{\gamma + 1}\right)^{\gamma/(\gamma-1)}. \tag{5}$$

$C_1$ and $C_2$ are constants that depend on the ideal gas constant $R$ and $\gamma$ and are given as:

$$C_1 = \sqrt{\frac{\gamma}{R}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}}, \quad C_2 = \sqrt{\left(\frac{2\gamma}{R(\gamma-1)}\right)} \tag{6}$$

Each chamber can operate in two different states: charging and discharging. During charging, $P_u$ is the supply pressure, $P_s$, while $P_d$ is the chamber's pressure, $P_{a/b}$. Conversely, during discharging, $P_u$ indicates the chamber pressure and $P_d$ indicates atmospheric pressure, $P_{atm}$. Considering the leakage flows ($\dot{m}_{a/bL}$) in each chamber, the mass flow rate for each is represented as:

$$\dot{m}_a = \begin{cases} A_m A_v \psi(P_s, P_a, C_{fin}) + \dot{m}_{aL} & , A_v \ge 0 \\ -A_m A_v \psi(P_a, P_{atm}, C_{fout}) + \dot{m}_{aL} & , A_v < 0 \end{cases} \tag{7}$$

$$\dot{m}_b = \begin{cases} -A_m A_v \psi(P_b, P_{atm}, C_{fout}) + \dot{m}_{bL} & , A_v \ge 0 \\ A_m A_v \psi(P_s, P_b, C_{fin}) + \dot{m}_{bL} & , A_v < 0 \end{cases} \tag{8}$$

where $\dot{m}_{aL}, \dot{m}_{bL}$ are:

$$\begin{cases} \dot{m}_{aL} = A_L \left(\psi(P_s, P_a, C_{f_{in}}) - \psi(P_a, P_{atm}, C_{f_{out}})\right) \\ \dot{m}_{bL} = A_L \left(\psi(P_s, P_b, C_{f_{in}}) - \psi(P_b, P_{atm}, C_{f_{out}})\right) \end{cases} \tag{9}$$

where $A_L$ is the cross-sectional leakage area and must be determined experimentally.

Taheri et al. [23] employed the same Enfield 5/3 proportional valve used in our study and developed a model relating the valve's aperture area, $A_v(z)$, to the spool position ($z$). In the present work, we will use the inverse of their equation as,

$$z(A_v) = \frac{-0.127 - 37.093A_v + 36.459A_v^2}{0.024 + 7.153A_v - 6.253A_v^2 - 0.748A_v^3} + 5.345 \tag{10}$$

### B. Load Dynamics with Friction Forces

The following equation can be obtained using Newton's second law:

$$M\ddot{x} = F_p - B_v\dot{x} - F_{S-C}(x, \dot{x}, P_a, P_b) \tag{11}$$

where $M$ represents the total mass, $\dot{x}$ velocity, $\ddot{x}$ acceleration, $B_v$ viscous friction coefficient, and $F_{S-C}(x)$ the combination of static and Coulomb friction forces. Various descriptions of friction models exist. For simplicity, this paper uses the following model:

$$F_{S-C}(x, \dot{x}, P_a, P_b) \triangleq \begin{cases} min\left((P_a A_a - P_b A_b), F_S(x)\right), & \dot{x} = 0 \\ F_C(x)\,\mathrm{sign}(\dot{x}), & \dot{x} \ne 0. \end{cases} \tag{12}$$

where $F_S(x)$ describes the static friction, usually dependent on piston position. $F_C(x) = F_C$ represents Coulomb friction, assumed constant. $F_C$ is generally smaller than $F_S(x)$.

## III. Sliding Mode Controller Design

Sliding-mode control can be used to follow the desired force in pneumatic systems that are modeled as a first-order control system with two variables, $P_a$ and $P_b$, as shown in Eq 2. SMC demonstrates strong capability in handling the uncertainties associated with the nonlinear dynamics of pneumatic actuators, as reported in [23]–[26]. The block diagram illustrating sliding mode control is depicted in Fig 2.

The system tracking error ($e$), and the sliding surface ($s$), for the valve aperture area controller are defined as:

$$e = F - F_d \quad \text{and} \quad s = e + \lambda \int e\,dt \tag{13}$$



Authorized licensed use limited to: Southern Methodist University. Downloaded on October 18,2026 at 21:43:04 UTC from IEEE Xplore. Restrictions apply.

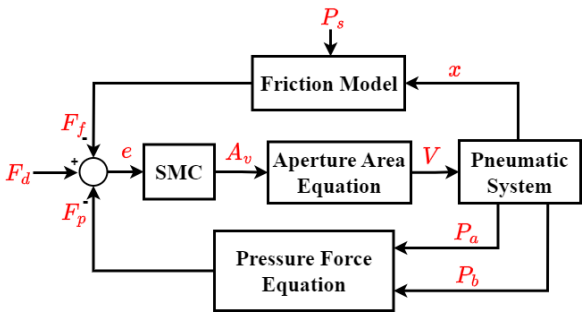



Fig. 2: Sliding mode controller with friction compensation block diagram.

where $F = F_p - F_f$ is the system output, $F_d$ is the desired force output of the pneumatic actuator, and $\lambda$ is a positive constant designating the weight of the integral error.

The equivalent control law ($u_e$) is determined by setting $\dot{s} = \dot{e} + \lambda e = 0$, and then solving for $A_v$. As mentioned, the pneumatic system has two operational states depending on the valve spool position. To streamline the process, it's effective to create two separate controllers—one for each state—and switch between them as necessary. When the spool position is positive ($A_v > 0$), air enters chamber A and leaves chamber B; the corresponding control law is given by Eq. 14. Conversely, when the spool position is negative ($A_v < 0$), air enters chamber B and exits chamber A, with the control law given by Eq. 15.

$$u_{eq_a} = \frac{\dot{F}_d + \dot{F}_f + \lambda(F_d - F) + \dot{x}\left(\frac{P_a A_a}{L_{ea}} + \frac{P_b A_b}{L_{eb}}\right) - A_L RT\left(\frac{\psi_a}{L_{ea}} - \frac{\psi_b}{L_{eb}}\right)}{A_m RT\left(\frac{\psi(P_s, P_a, C_{fin})}{L_{ea}} + \frac{\psi(P_b, P_{atm}, C_{fou})}{L_{eb}}\right)} \tag{14}$$

$$u_{eq_b} = -\frac{\dot{F}_d + \dot{F}_f + \lambda(F_d - F) + \dot{x}\left(\frac{P_a A_a}{L_{ea}} + \frac{P_b A_b}{L_{eb}}\right) - A_L RT\left(\frac{\psi_a}{L_{ea}} - \frac{\psi_b}{L_{eb}}\right)}{A_m RT\left(\frac{\psi(P_s, P_{atm}, C_{f_{in}})}{L_{ea}} + \frac{\psi(P_s, P_b, C_{f_{in}})}{L_{eb}}\right)} \tag{15}$$

where $\psi_a$ and $\psi_a$ are the net normalized mass flows into chambers A and B, respectively, given by Eq. 16, and $L_{ea}$ and $L_{eb}$ are new variables defined by Eq. 17.

$$\begin{aligned} \psi_a &= \psi(P_s, P_a, C_{f_{in}}) - \psi(P_a, P_{atm}, C_{f_{out}}) \\ \psi_b &= \psi(P_s, P_b, C_{f_{in}}) - \psi(P_b, P_{atm}, C_{f_{out}}) \end{aligned} \tag{16}$$

$$L_{ea} = \frac{v_{0a}}{A_a} + \frac{L}{2} + x, \qquad L_{eb} = \frac{v_{0b}}{A_b} + \frac{L}{2} - x \tag{17}$$

To reduce excessive chattering, a boundary layer is added into the sliding surface, smoothing the discontinuity caused by the sign-dependent robustness term. The system's configuration determines which of two outcomes occurs:

$$u_a = u_{eqa} - \kappa\, sat\left(\frac{e}{\phi}\right) \tag{18}$$

$$u_b = u_{eqb} + \kappa\, sat\left(\frac{e}{\phi}\right) \tag{19}$$

where $\kappa$ is the robustness term that defines the weight of the uncertainty component relative to the rest of the controller, $\phi$ defines the thickness of the boundary layer, and:

$$sat(y) = \begin{cases} y & , |y| \le 1 \\ sgn(y) & , |y| > 1 \end{cases} \tag{20}$$

## IV. Experimental Setup and Results

Figure 3 illustrates the experimental framework for both static friction modeling and SMC implementation. Both experiments include an Enfield Technologies LS-V05s Proportional Pneumatic Control Valve, driven by an Enfield Technologies LS-C21 Analog Valve Driver, which controls the compressed air flow to a Bimba Round Body 5/16" Bore Double-Acting Air Cylinder. The actuator is equipped with two Honeywell 100PAAA5 TruStability Board-Mount Pressure Sensors that monitor the pressure in each chamber, as well as an additional sensor to measure the supply pressure. Moreover, a Honeywell MLT-38000104 linear displacement sensor tracks the piston's displacement. In the SMC implementation, an MLP-10 strain gauge load cell is attached to the tip of the rod to measure the generated force.

The code was developed with the LabVIEW programming language on an NI cRIO-9024 real-time system that includes an FPGA DAQ module. Table I presents the parameters used in the experiments. Some parameters are defined by the design or environmental conditions, while others require experimental determination.

### A. Friction Identification

This experiment is designed to measure the friction force and develop a related mathematical model. The piston rod

TABLE I: System Parameters: Cylinder, Valve, Environmental, Experimental, and Controller Values.

| Parameter | Value | Description |
|---|---|---|
| **Cylinder and Valve Parameters** | | |
| $L$ | 38.1 mm | Stroke length of cylinder |
| $D_p$ | 7.94 mm | Piston inner diameter |
| $D_r$ | 3.18 mm | Cylinder rod diameter |
| $D_t$ | 2.38 mm | Pneumatic tube inner diameter |
| $L_{ta}$ | 840 mm | Tube length to chamber $A$ |
| $L_{tb}$ | 840 mm | Tube length to chamber $B$ |
| $A_m$ | 5 mm$^2$ | Valve cross-sectional area |
| **Environmental Conditions** | | |
| $R$ | 287.05 J/kg·K | Ideal gas constant |
| $T$ | 295 K | Ambient temperature |
| $P_{atm}$ | 101.325 kPa | Atmospheric pressure |
| $\gamma$ | 1.4 | Specific heat ratio for air |
| $P_{cr}$ | 0.528 | Critical pressure ratio |
| $C_1$ | $1.278 \times 10^{-3}$ | Choked flow coefficient |
| $C_2$ | $4.938 \times 10^{-3}$ | Subsonic flow coefficient |
| **Experimental Parameters** | | |
| $A_L$ | 0.05 mm$^2$ | Leak cross-sectional area |
| $C_{fin}$ | 4.5 | Flow coefficient into chamber |
| $C_{fout}$ | 3 | Flow coefficient out of chamber |
| **Controller Parameters** | | |
| $\lambda$ | 38 | Integral coefficient |
| $\kappa$ | 0.02 | Robustness term |
| $\phi$ | 2 N | Boundary layer thickness |



Authorized licensed use limited to: Southern Methodist University. Downloaded on October 18,2026 at 21:43:04 UTC from IEEE Xplore. Restrictions apply.

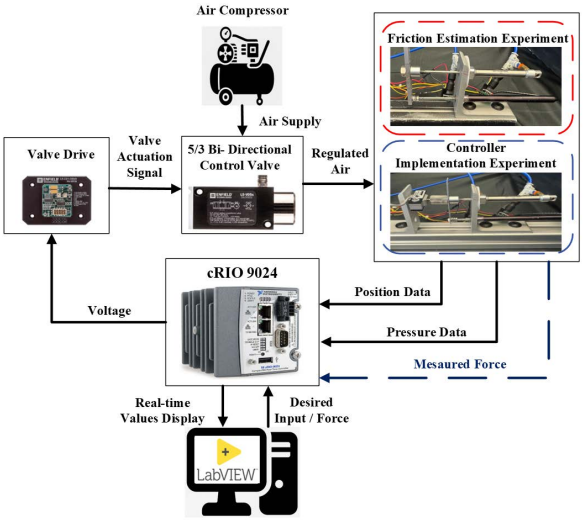



Fig. 3: Friction estimation and SMC implementation experiment framework.

is only attached to a linear position sensor and operates without any external load. Starting from rest at a specific position, the valve is slightly opened to introduce a slow airflow into chamber A, gradually increasing the pressure and pushing the piston outward. The pressures in both chambers, the measured piston position, and velocity (obtained by differentiating the position data and filtering with a 15 Hz low-pass filter at a 1 kHz sampling rate) were recorded. The force required to move the cylinder was then calculated using chamber pressures by Eq. 1. An example of signals recorded during the experiment is illustrated in Fig. 4.

The experiment was conducted for 17 piston positions spanning the entire actuator stroke, at four pressure levels. Each measurement was repeated three times, and the results were averaged. The same process was repeated for piston retraction by raising the pressure in chamber $B$.

Figure 5 presents the static friction force at different piston positions under four supply pressures during piston extension and retraction. The results indicate that the friction force increases as the supply pressure increases, and it decreases as the piston position is changed from fully retracted to fully

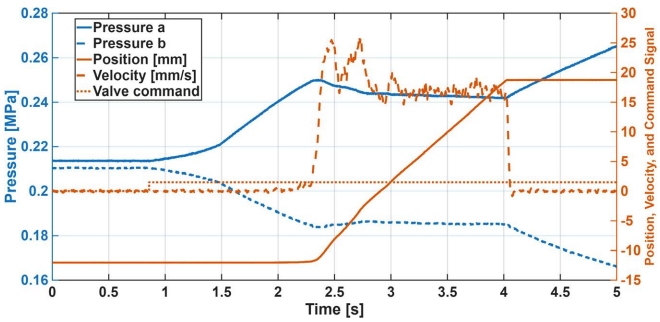
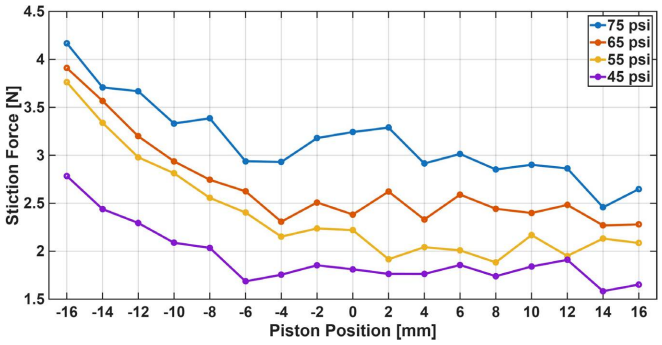
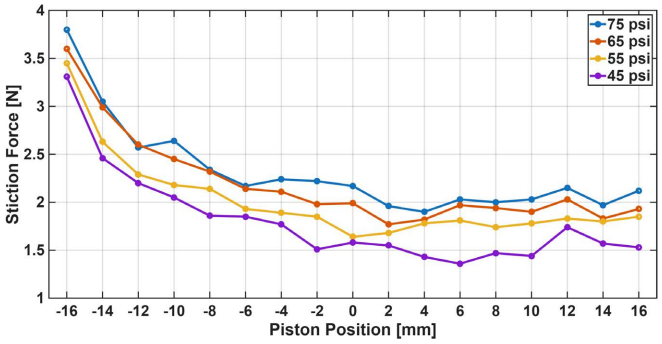



Fig. 4: Chamber pressures, piston position, and piston velocity measured and recorded during the experiment for extension motion ($P_s = 0.4482$ MPa, $x = -12$ mm).




(a)




(b)

Fig. 5: Measured friction force versus piston position under multiple working pressures during: (a) extension and (b) retraction.

extended. Additionally, the friction force during extension is higher than during retraction, and these results are consistent with those in [27], which show that friction depends on the direction of motion and that friction behavior is often non-symmetrical. Figure 6 illustrates the distribution of measured friction forces at various supply pressures, clearly showing that static friction increases as pressure rises.

Based on the analysis of the experimental data for friction–pressure and friction–position relationships, two candidate models are proposed:

$$F_{f1}(x,P_s) = (b_1 + b_2 P_s)e^{b_3 x} + (b_4 + b_5 P_s)e^{b_6 x} + b_7 \quad (21)$$

$$F_{f2}(x,P_s) = b_1 e^{(b_2 + b_3 P_s)x} + b_4 + b_5 P_s \quad (22)$$

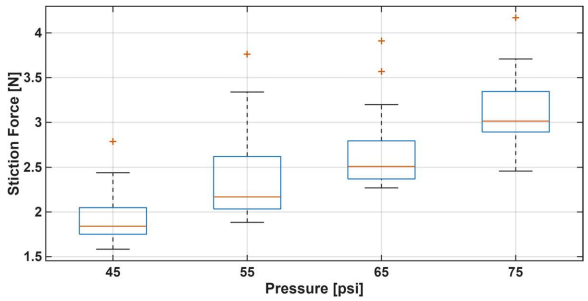



Fig. 6: Box plot of friction force across supply pressures during extension motion.



Authorized licensed use limited to: Southern Methodist University. Downloaded on October 18,2026 at 21:43:04 UTC from IEEE Xplore. Restrictions apply.

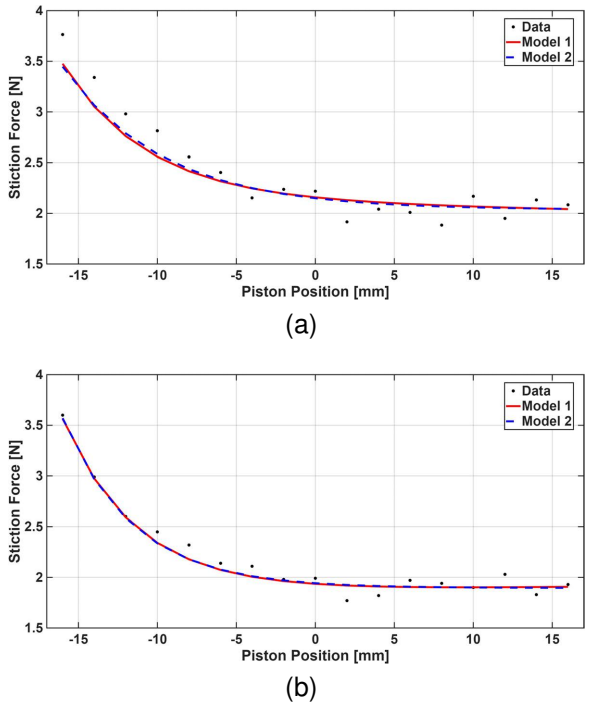



Fig. 7: Measured friction force versus piston position at 65 psi with fitted curves during: (a) extension and (b) retraction.

Two separate friction functions were developed, one for extension and one for retraction. The models' parameters were estimated using MATLAB's nonlinear regression function, `fitnlm`, as shown in Table II. Both models demonstrate high accuracy, with $R^2$ values exceeding 0.93 and RMSE values below 0.16 in both extension and retraction phases. Figure 7 shows the experimental data and the fitted curves using both models during piston extension and retraction at a pressure supply of 65 psi. Table III shows goodness-of-fit metrics for both proposed models. Model 2 performs similarly to Model 1 but is more compact, with fewer parameters, and is therefore adopted for implementation in the controller.

### *B. Force Controller with Friction Compensation*

This experiment aims to demonstrate the SMC's ability to generate the desired force with friction compensation while the piston rod remains stationary. In this configuration, both

TABLE II: Parameter values of the proposed friction models for extension (Ext.) and retraction (Ret.).

| Parameter | Model 1 | | Model 2 | |
|---|---|---|---|---|
| | Ext. | Ret. | Ext. | Ret. |
| $b_1$ | 0.0609 | 0.0569 | 0.1140 | 0.0478 |
| $b_2$ | -0.0055 | 0.0003 | -0.1582 | -0.2222 |
| $b_3$ | -0.1893 | -0.2126 | -0.0021 | -0.0002 |
| $b_4$ | 0.5074 | 281.77 | 2.2288 | 1.8095 |
| $b_5$ | 0.4606 | 0.1917 | 0.4358 | 0.1922 |
| $b_6$ | -0.0128 | 0 | — | — |
| $b_7$ | 1.7921 | -279.98 | — | — |

TABLE III: Goodness-of-fit metrics for the proposed models

| Metric | Model 1 | | Model 2 | |
|---|---|---|---|---|
| | Extension | Retraction | Extension | Retraction |
| RMSE | 0.1585 | 0.0988 | 0.1612 | 0.0982 |
| $R^2$ | 0.9406 | 0.9644 | 0.9366 | 0.9643 |

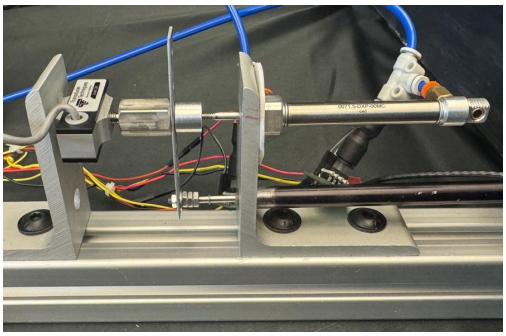


Fig. 8: Experimental setup for controller implementation, showing the cylinder, linear sensor, and load cell.

the load cell and the cylinder are fixed to a rail, as shown in Fig. 8.

The SMC with integrated friction compensation decreases steady-state force tracking errors compared to the uncompensated controller and reduces the time delay (Fig. 9 and Table IV).

TABLE IV: Comparison of friction-compensated and uncompensated SMC performance at 16 N and 1 Hz and 3 Hz.

| Metric | 1 Hz | | 3 Hz | |
|---|---|---|---|---|
| | Comp. | Uncomp. | Comp. | Uncomp. |
| Amplitude [N] | 15.31 | 10.91 | 16.85 | 12.09 |
| Relative error [%] | 4.29 | 31.79 | 5.3 | 24.43 |
| Time delay [ms] | 45 | 81 | 17 | 26 |

## V. Conclusions

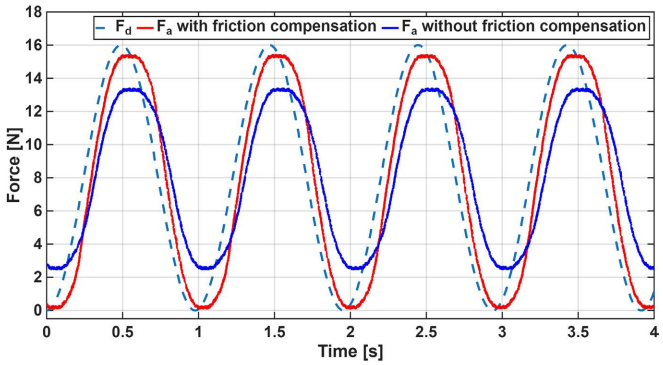
This paper investigates the static friction of a small pneumatic cylinder, where its effect is significant, reaching more than 30% of the full-scale value. Two models for static friction as a function of both pressure and piston position have been evaluated. The simplest model with fewer parameters was shown to effectively estimate the friction, with $R^2$ values over 0.93 and RMSE below 0.17 during both extension and retraction.

A Sliding Mode Control was developed and tested both with and without friction compensation to evaluate the system's ability to produce the desired force. Results indicate that SMC with integrated friction compensation reduces the tracking error for the force amplitude by more than 78%. In addition, the trajectory tracking time delay was reduced by more than 36%.

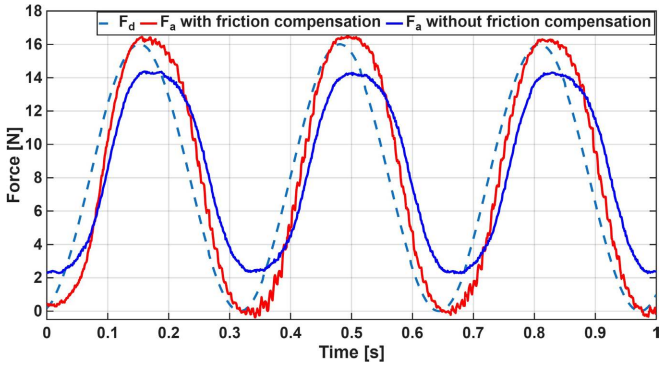
Future work includes broadening the experimental and modeling scope to include both static and dynamic friction, as well as implementing and testing the controller with the



Authorized licensed use limited to: Southern Methodist University. Downloaded on October 18,2026 at 21:43:04 UTC from IEEE Xplore. Restrictions apply.




(a)




(b)

Fig. 9: Comparison of SMC with integrated friction compensation and uncompensated controller in tracking a force reference: (a) $F_d = 16$ N, 1 Hz haversine wave, and (b) $F_d = 16$ N, 3 Hz haversine wave

full dynamic model of friction across different input types with varying magnitudes and frequencies.

## REFERENCES


[1] M. K. O'malley and A. Gupta, "Haptic interfaces," *HCI beyond the GUI: Design for Haptic, Speech, Olfactory, and other nontraditional Interfaces*, pp. 25–64, 2008, publisher: San Francisco: Morgan Kaufmann Publishers Inc.

[2] V. Z. Pérez Ariza and M. Santís-Chaves, "Haptic interfaces: kinesthetic vs. tactile systems," *Revista EIA*, no. 26, pp. 13–29, 2016, publisher: Escuela de ingenieria de Antioquia.

[3] M. Kang, C.-G. Gang, S.-K. Ryu, H.-J. Kim, D.-Y. Jeon, and S. Pyo, "Haptic interface with multimodal tactile sensing and feedback for human–robot interaction," *Micro and Nano Systems Letters*, vol. 12, no. 1, p. 9, Mar. 2024.

[4] A. Gani, O. Pickering, C. Ellis, O. Sabri, and P. Pucher, "Impact of haptic feedback on surgical training outcomes: a randomised controlled trial of haptic versus non-haptic immersive virtual reality training," *Annals of Medicine and Surgery*, vol. 83, p. 104734, 2022, publisher: Elsevier.

[5] N. Enayati, E. De Momi, and G. Ferrigno, "Haptics in robot-assisted surgery: Challenges and benefits," *IEEE reviews in biomedical engineering*, vol. 9, pp. 49–65, 2016, publisher: IEEE.

[6] Y. S. Kim, J. Lee, S. Lee, and M. Kim, "A force reflected exoskeleton-type masterarm for human-robot interaction," *IEEE Transactions on Systems, Man, and Cybernetics-Part A: Systems and Humans*, vol. 35, no. 2, pp. 198–212, 2005, publisher: IEEE.

[7] E. Al Khatib, X. Zhang, M. J. Kim, and Y. Hurmuzlu, "Teleoperation control scheme for magnetically actuated microrobots with haptic guidance," *Journal of Micro-Bio Robotics*, vol. 16, no. 2, pp. 161–171, 2020, publisher: Springer.

[8] S. Demain, C. D. Metcalf, G. V. Merrett, D. Zheng, and S. Cunningham, "A narrative review on haptic devices: relating the physiology and psychophysical properties of the hand to devices for rehabilitation in central nervous system disorders," *Disability and Rehabilitation: Assistive Technology*, vol. 8, no. 3, pp. 181–189, 2013, publisher: Taylor & Francis.

[9] D. A. Lawrence and J. D. Chapel, "Performance trade-offs for hand controller design," in *Proceedings of the 1994 IEEE International Conference on Robotics and Automation*. IEEE, 1994, pp. 3211–3216.

[10] Y. Hurmuzlu, A. Ephanov, and D. Stoianovici, "Effect of a pneumatically driven haptic interface on the perceptional capabilities of human operators," *Presence*, vol. 7, no. 3, pp. 290–307, 1998, publisher: MIT Press.

[11] A. Adilkhanov, M. Rubagotti, and Z. Kappassov, "Haptic devices: Wearability-based taxonomy and literature review," *IEEE Access*, vol. 10, pp. 91 923–91 947, 2022, publisher: IEEE.

[12] C. Pacchierotti, S. Sinclair, M. Solazzi, A. Frisoli, V. Hayward, and D. Prattichizzo, "Wearable haptic systems for the fingertip and the hand: taxonomy, review, and perspectives," *IEEE transactions on haptics*, vol. 10, no. 4, pp. 580–600, 2017, publisher: IEEE.

[13] P. Ben-Tzvi and Z. Ma, "Sensing and force-feedback exoskeleton (safe) robotic glove," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 23, no. 6, pp. 992–1002, 2014, publisher: IEEE.

[14] M. Bouzit, G. Burdea, G. Popescu, and R. Boian, "The Rutgers Master II-new design force-feedback glove," *IEEE/ASME Transactions on mechatronics*, vol. 7, no. 2, pp. 256–263, 2002, publisher: IEEE.

[15] A. Yacoub, E. Richer, and Y. Hurmuzlu, "Comparative Analysis of Fingertip Location for the SMU Haptic Glove by OptiTrack Cameras and Embedded Position Sensors," *IFAC-PapersOnLine*, vol. 59, no. 30, pp. 203–208, 2025, publisher: Elsevier.

[16] D. Serrano, D. Copaci, J. Arias, L. E. Moreno, and D. Blanco, "SMA-based soft exo-glove," *IEEE Robotics and Automation Letters*, vol. 8, no. 9, pp. 5448–5455, 2023, publisher: IEEE.

[17] X. Tran, H. Dao, and K. Tran, "A new mathematical model of friction for pneumatic cylinders," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, vol. 230, no. 14, pp. 2399–2412, Aug. 2016.

[18] A. H. Zamanian and E. Richer, "Identification and compensation of cogging and friction forces in tubular permanent magnet linear motors," in *Dynamic Systems and Control Conference*, vol. 58288. American Society of Mechanical Engineers, 2017, p. V002T04A003.

[19] X. B. Tran, V. L. Nguyen, and K. D. Tran, "Effects of friction models on simulation of pneumatic cylinder," pp. 517–528, 2019, issue: 2 Publisher: Copernicus Publications Göttingen, Germany.

[20] A. H. Zamanian and E. Richer, "Adaptive notch filter for pathological tremor suppression using permanent magnet linear motor," *Mechatronics*, vol. 63, p. 102273, 2019, publisher: Elsevier.

[21] J. Wang, J. D. Wang, N. Daw, and Q. H. Wu, "Identification of pneumatic cylinder friction parameters using genetic algorithms," pp. 100–107, 2004, issue: 1 Publisher: IEEE.

[22] E. Richer and Y. Hurmuzlu, "A high performance pneumatic force actuator system: Part I—Nonlinear mathematical model," *J. dyn. sys., meas., control*, vol. 122, no. 3, pp. 416–425, 2000.

[23] B. Taheri, D. Case, and E. Richer, "Force and stiffness backstepping-sliding mode controller for pneumatic cylinders," *IEEE/ASME Transactions on Mechatronics*, vol. 19, no. 6, pp. 1799–1809, 2014, publisher: IEEE.

[24] E. Richer and Y. Hurmuzlu, "A high performance pneumatic force actuator system: part II—nonlinear controller design," *J. Dyn. Sys., Meas., Control*, vol. 122, no. 3, pp. 426–434, 2000.

[25] X. Shen and M. Goldfarb, "Simultaneous force and stiffness control of a pneumatic actuator," 2007.

[26] E. Richer, M. Galla, and Y. Hurmuzlu, "Directional Force Feedback for a 3 DOF Pneumatic Haptic Finger," *IFAC-PapersOnLine*, vol. 56, no. 3, pp. 295–300, 2023, publisher: Elsevier.

[27] G. Belforte, A. M. Bertetto, and L. Mazza, "Test rig for friction force measurements in pneumatic components and seals," *Proceedings of the Institution of Mechanical Engineers, Part J: Journal of Engineering Tribology*, vol. 227, no. 1, pp. 43–59, Jan. 2013.




Authorized licensed use limited to: Southern Methodist University. Downloaded on October 18,2026 at 21:43:04 UTC from IEEE Xplore. Restrictions apply.